

EECS 489

Computer Networks

Winter 2025

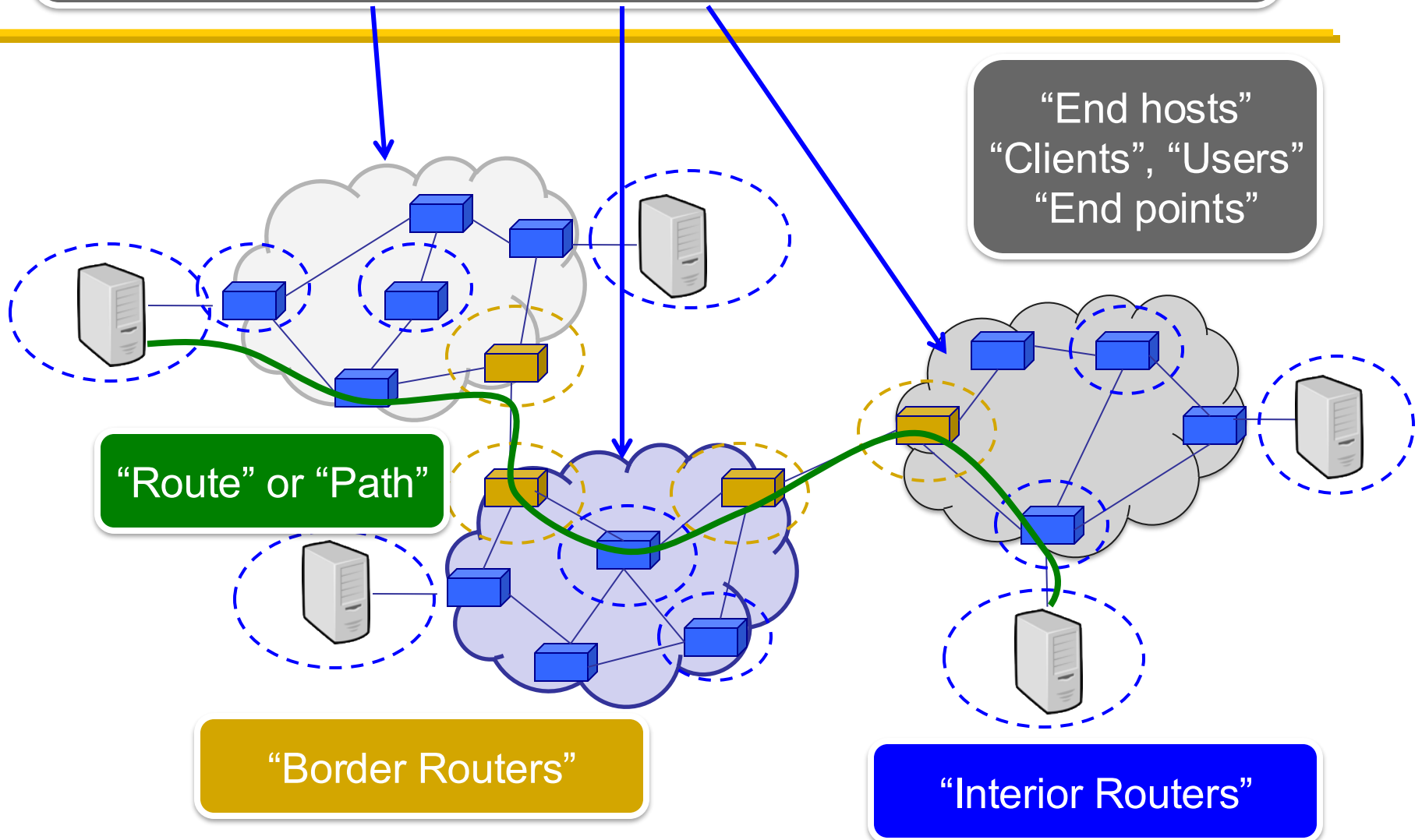
Mosharaf Chowdhury

Material with thanks to Aditya Akella, Sugih Jamin, Philip Levis, Sylvia Ratnasamy, Peter Steenkiste, and many other colleagues.

Agenda

- Inter-domain-routing

“Autonomous System (AS)” or “Domain”
Region of a network under a single administrative entity



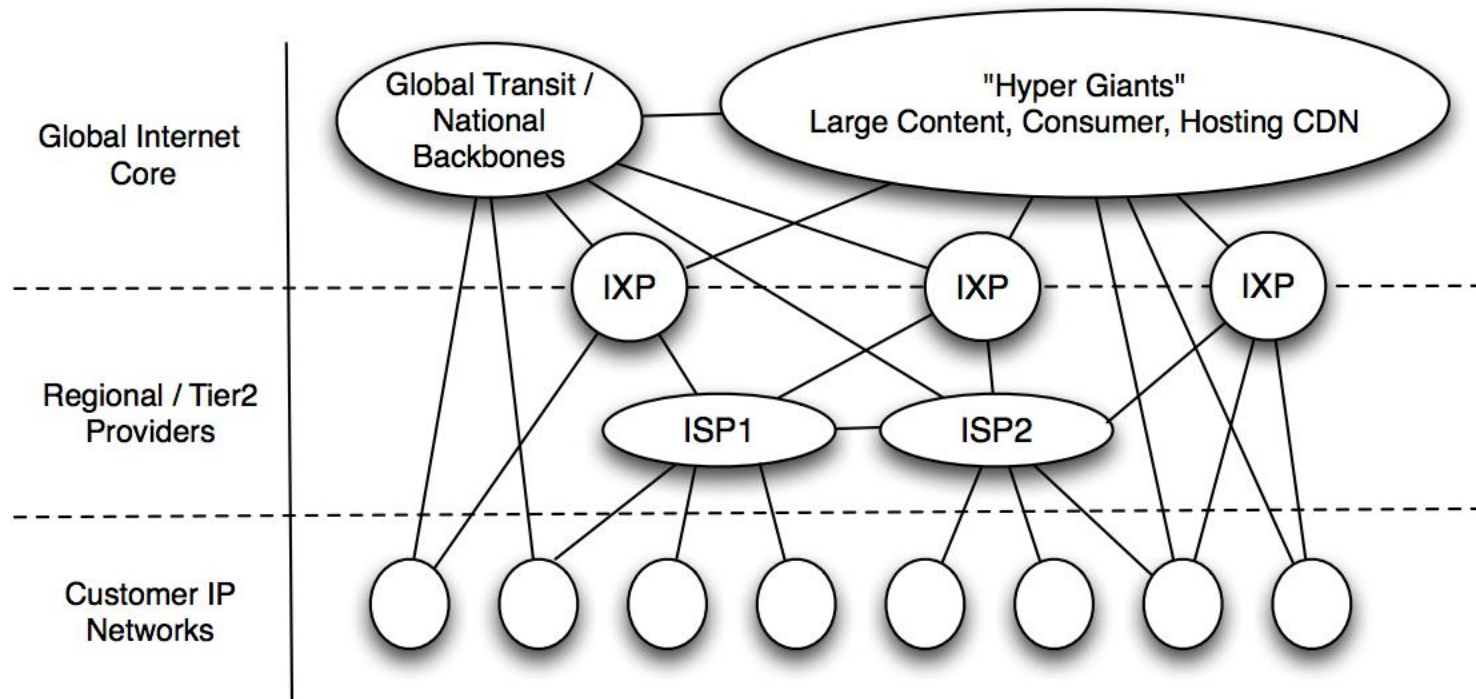
Autonomous systems (AS)

- An AS is a network under a single administrative control
 - Currently, ~77,000 IPv4 ASes & 34,750+ IPv6 ASes
 - » Source: <https://radar.cloudflare.com/routing>
- ASes are sometimes called “domains”
- Each AS is assigned a unique identifier (ASN)
 - E.g., [University of Michigan owns ASNs 177 to 180](#)

AS-level Internet

- Used to be a large graph of ASes
 - In 2007, half of the Internet's total traffic came from ~2000 ASes
- It's consolidating since then
 - In 2009, the largest 150 ASes contributed to half the traffic

AS-level Internet



Internet Inter-Domain Traffic, SIGCOMM, 2010

AS-level Internet Today

- By 2019, half the Internet traffic came from only five hypergiants such as Google, Netflix, Meta, Akamai
 - Seven Years in the Life of Hypergiants' Off-nets, SIGCOMM, 2021

“Intra-domain” routing: Within an AS

- Link-State (e.g., OSPF) and Distance-Vector (e.g., RIP)
- Primary focus
 - Finding least-cost paths
 - Fast convergence

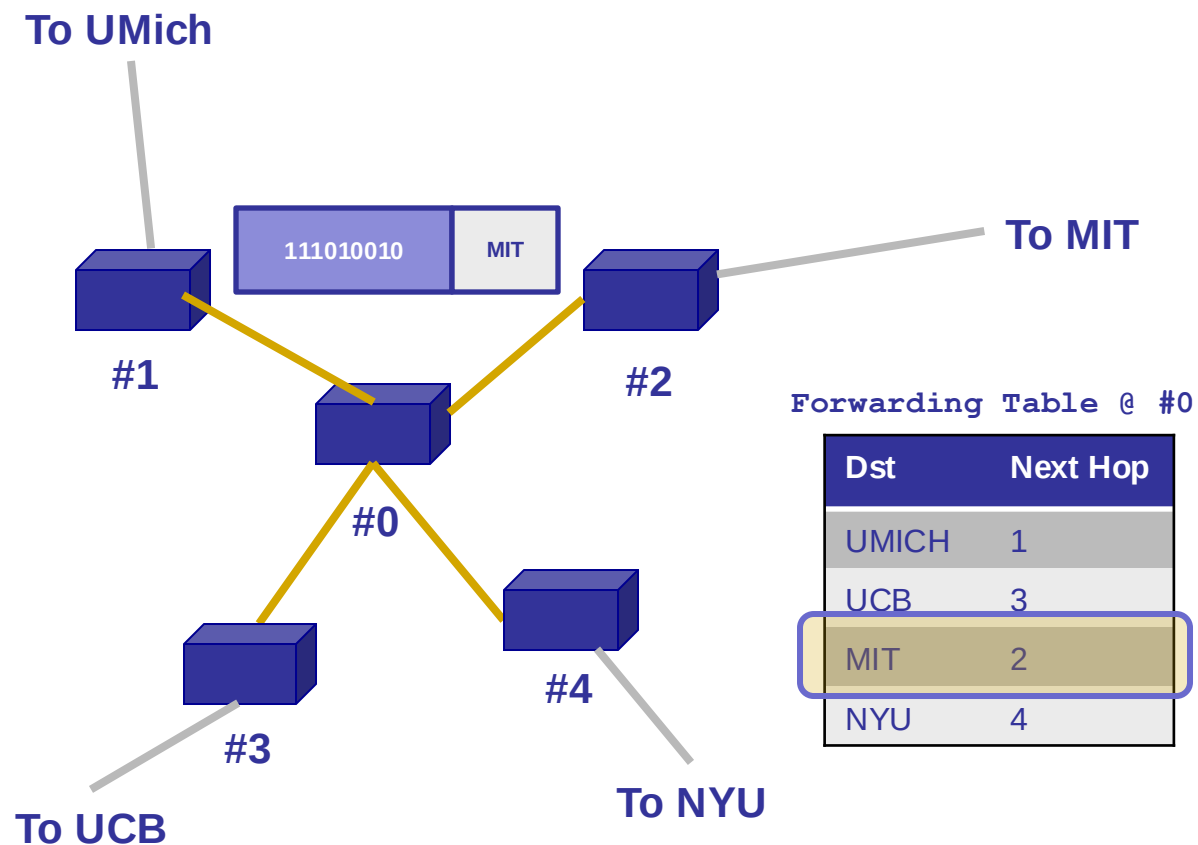
“Inter-domain” routing: Between ASes

- Two key challenges
 - Scaling
 - Administrative structure
 - » Issues of autonomy, policy, privacy

Recall: Addressing (so far)

- Each host has a unique ID
- No particular structure to those IDs

Recall: Forwarding



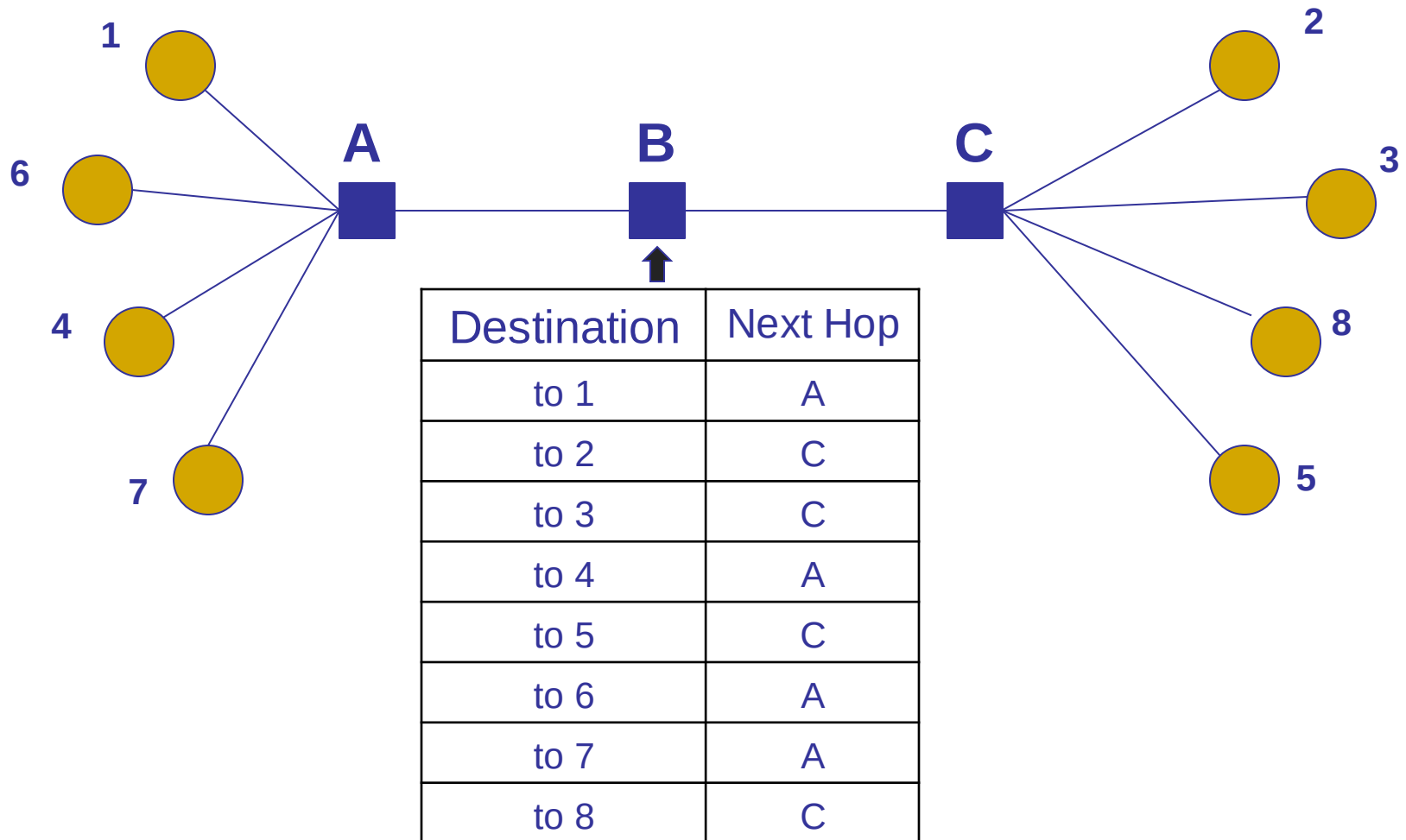
Two key challenges

- ▣ Scaling
- ▣ Administrative structure
 - Issues of autonomy, policy, privacy

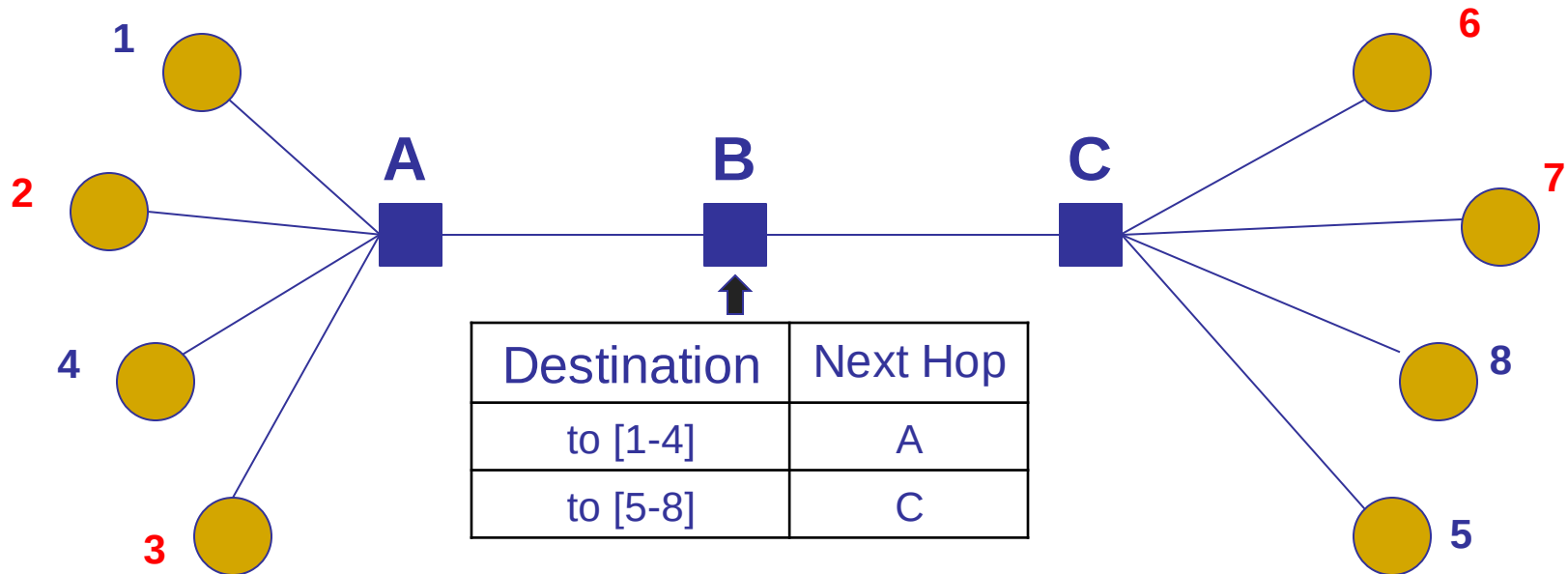
Scaling

- ❑ A router must be able to reach any destination
 - Given packet's destination address, lookup next hop
- ❑ Naive: Have an entry for each destination
 - There would be over 10^8 entries!
 - **AND** routing updates per destination!
- ❑ **How can we improve scalability?**
 - We have already seen an example: **longest-prefix matching**

A smaller table at node B?



Re-number the end-systems?



- Careful address assignment → can *aggregate* multiple addresses into one range → scalability!
- Akin to reducing the number of destinations

Scaling

- ❑ A router must be able to reach any destination
- ❑ Naive: Have an entry for each destination
- ❑ Better: Have an entry for a range of addresses
 - Can't do this if addresses are assigned randomly!
 - How addresses are allocated will matter!
- ❑ Host addressing is key to scaling

Two key challenges

- ▣ Scaling
- ▣ Administrative structure
 - Issues of autonomy, policy, privacy

Administrative structure shapes inter-domain routing

- ASes want freedom in picking routes
 - “My traffic can’t be carried over my competitor’s network”
 - “I don’t want to carry A’s traffic through my network”
 - Not expressible as Internet-wide “least cost”
- ASes want autonomy
 - Want to choose their own internal routing protocol
 - Want to choose their own policy
- ASes want privacy
 - Choice of network topology, routing policies, etc.

Choice of routing algorithm

□ Link-state

- No privacy – broadcasts all network information
- Limited autonomy – needs agreement on metric, algo

□ Distance-vector is a decent starting point

- Per-destination updates give some control
- BUT wasn't designed to implement policy
- AND is vulnerable to loops

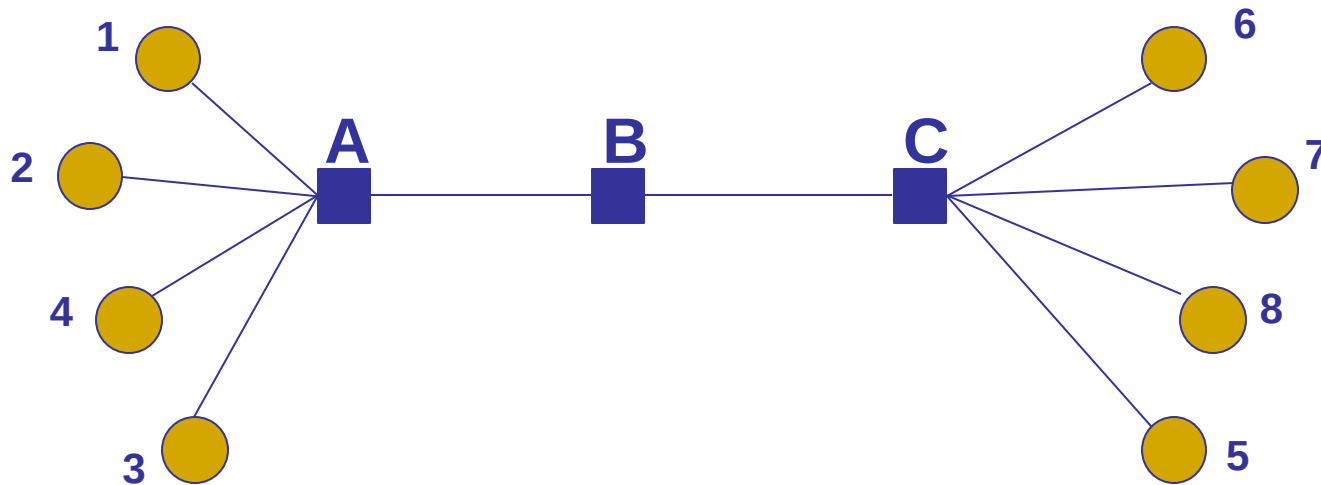
□ The “Border Gateway Protocol” (BGP) extends distance-vector ideas to accommodate policy

IP ADDRESSING

Goal of addressing: Scalable routing

- State: Small forwarding tables at routers
 - Much less than the number of hosts
- Churn: Limited rate of change in routing tables
- Ability to aggregate addresses is crucial for both

Aggregation works if...



- ? Groups of destinations reached via the same path
- ? These groups are assigned contiguous addresses
- ? These groups are relatively stable
- ? Few enough groups to make forwarding easy

IP addressing is hierarchical

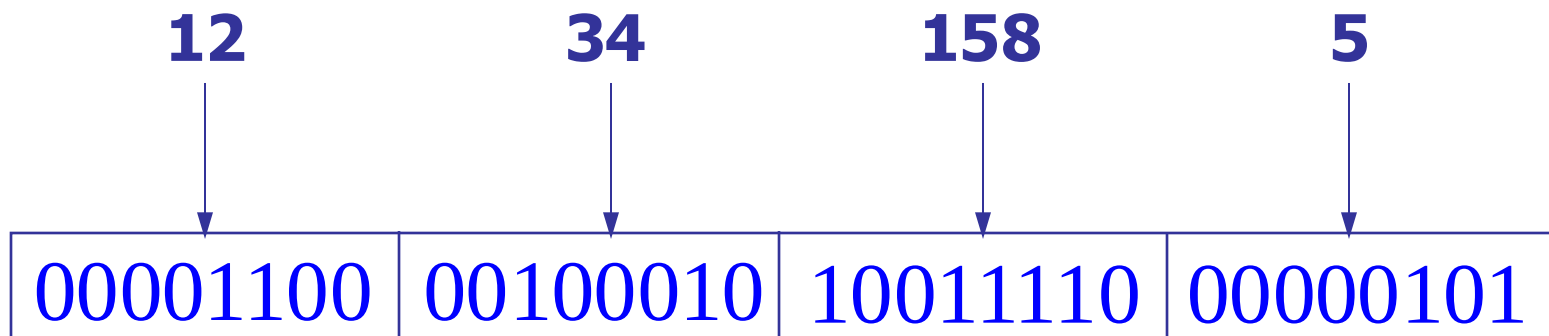
- Hierarchical address structure
- Hierarchical address allocation
- Hierarchical addresses and routing scalability

IP addresses (IPv4)

- Unique 32-bit number associated with a host

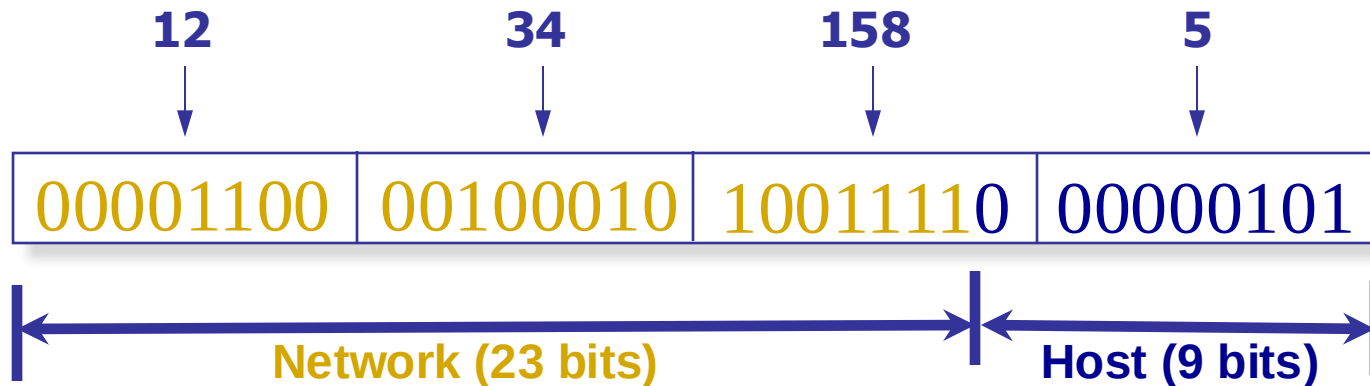
00001100 00100010 10011110 00000101

- Represented with the “dotted-decimal” notation
 - e.g., 12.34.158.5



Hierarchy in IP addressing

- 32 bits are partitioned into a prefix and suffix components
- Prefix is the **network** component; suffix is the **host** component



- Inter-domain routing operates on network prefix

CIDR: Classless inter-domain routing

- ▣ Flexible division between network and host addresses
- ▣ Offers a better tradeoff between size of the routing table and efficient use of the IP address space

CIDR example

- Suppose a network has 50 computers
 - Allocate 6 bits for host addresses ($2^5 < 50 < 2^6$)
 - Remaining $32 - 6 = 26$ bits as network prefix
- Flexible boundary means the boundary must be explicitly specified with the network address!
 - Informally, “slash 26” $\rightarrow$ 128.23.9/26
 - Formally, prefix represented with a 32-bit mask: 255.255.255.192, where all network prefix bits set to “1” and host suffix bits to “0”
 - Also known as **subnet mask** (a group of machines with the same prefix are in the same subnet)

Before CIDR: Classful addressing

□ Three classes

- 8-bit network prefix (Class A),
- 16-bit network prefix (Class B), or
- 24-bit network prefix (Class C)

□ Example: an organization needs 500 addresses.

- A single class C address is not enough (<500 hosts)
- Instead, a class B address is allocated (~65K hosts)
 - » Huge waste!

5-MINUTE BREAK!

IP addressing is hierarchical

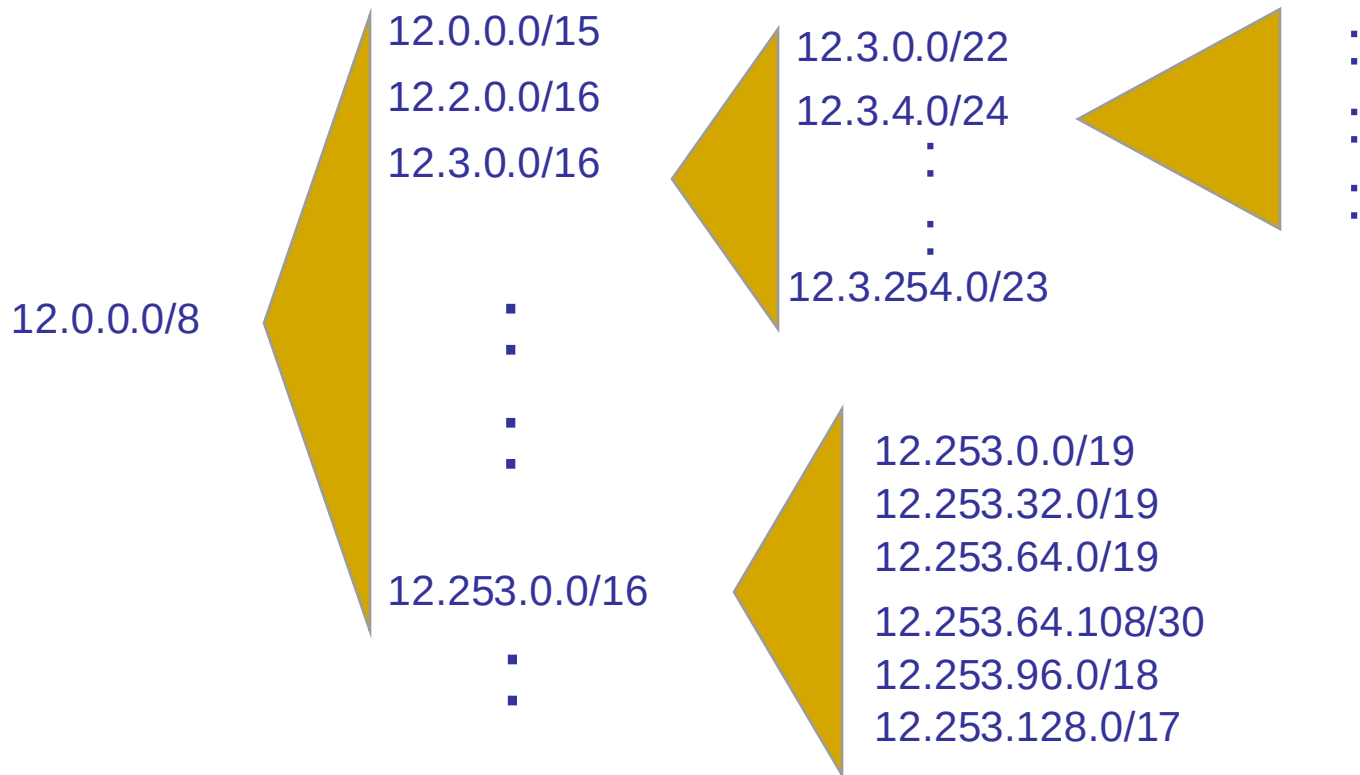
- Hierarchical address structure
- Hierarchical address allocation
- Hierarchical addresses and routing scalability

Allocation done hierarchically

- ❑ Internet Corporation for Assigned Names and Numbers (ICANN) gives large blocks to...
- ❑ Regional Internet Registries, such as the American Registry for Internet Names (ARIN), which give blocks to...
- ❑ Large institutions (ISPs), which give addresses to...
- ❑ Individuals and smaller institutions
- ❑ FAKE Example:
 - ICANN → ARIN → AT&T → UMich → EECS

CIDR: Addresses allocated in contiguous prefix chunks

- Recursively break down chunks as get closer to host



FAKE example in more detail

- ❑ ICANN gave ARIN several /8s
- ❑ ARIN gave AT&T one /8, **12.0/8**
 - Network Prefix: **00001100**
- ❑ AT&T gave UMich a /16, **12.34/16**
 - Network Prefix: **0000110000100010**
- ❑ UMich gave EECS a /24, **12.34.56/24**
 - Network Prefix: **000011000010001000111000**
- ❑ EECS gave me specific address **12.34.56.78**
 - Address: **00001100001000100011100001001110**

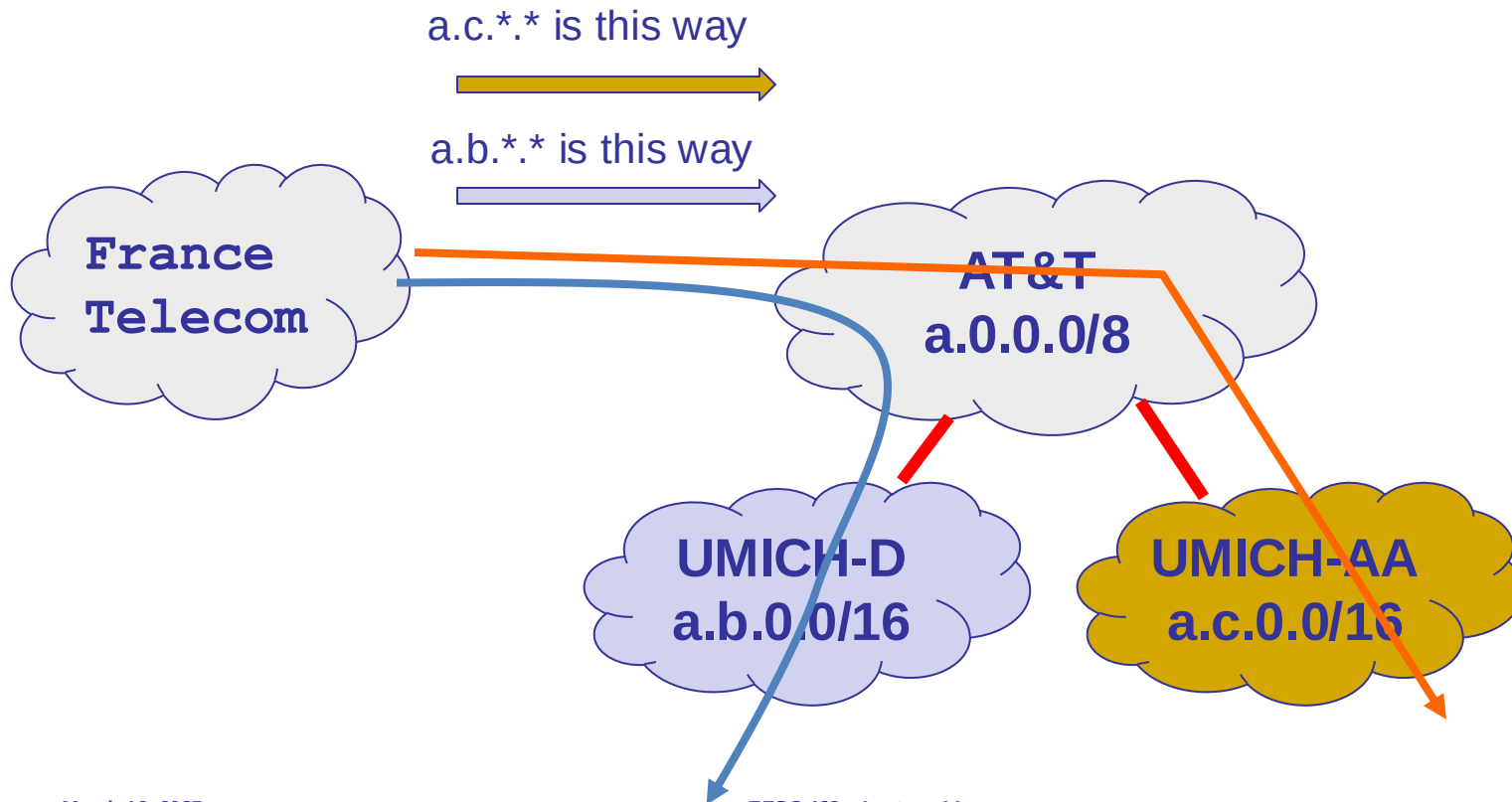
IP addressing is hierarchical

- Hierarchical address structure
- Hierarchical address allocation
- Hierarchical addresses and routing scalability

IP addressing → Scalable routing?

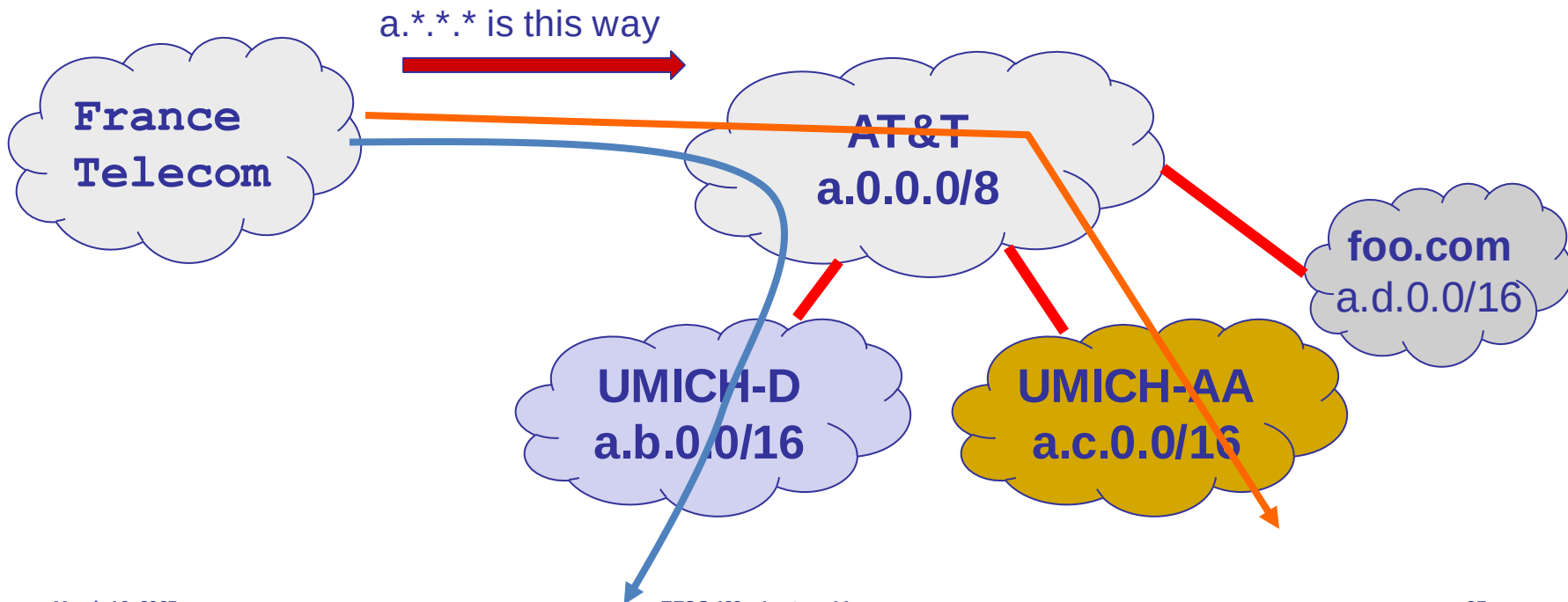
- Hierarchical address allocation only helps routing scalability if allocation matches topological hierarchy

IP addressing → Scalable routing?



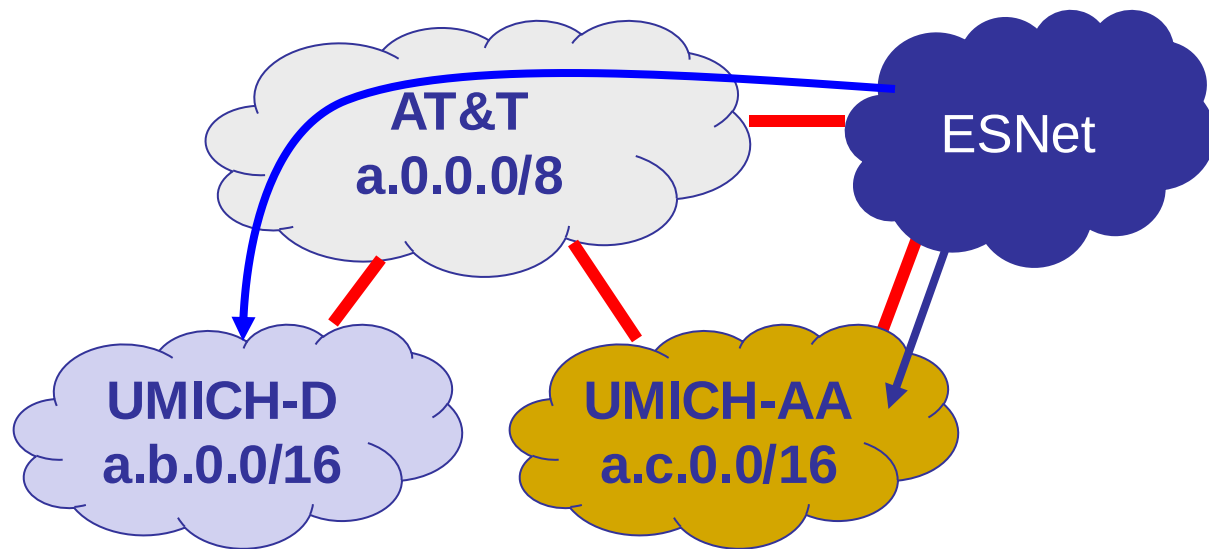
IP addressing → Scalable routing?

Can add new hosts/networks without updating the routing entries at France Telecom



IP addressing → Scalable routing?

ESNet must maintain routing entries for both $a.*.*$ and $a.c.*.*$



IP addressing → Scalable routing?

- ❑ Hierarchical address allocation only helps routing scalability if allocation matches topological hierarchy
- ❑ May not be able to aggregate addresses for “multi-homed” networks
 - A multi-homed network is connected to more than one ASes for fault-tolerance, load balancing, etc.

BGP: BORDER GATEWAY PROTOCOL

BGP (Today)

- The role of policy
 - What we mean by it
 - Why we need it

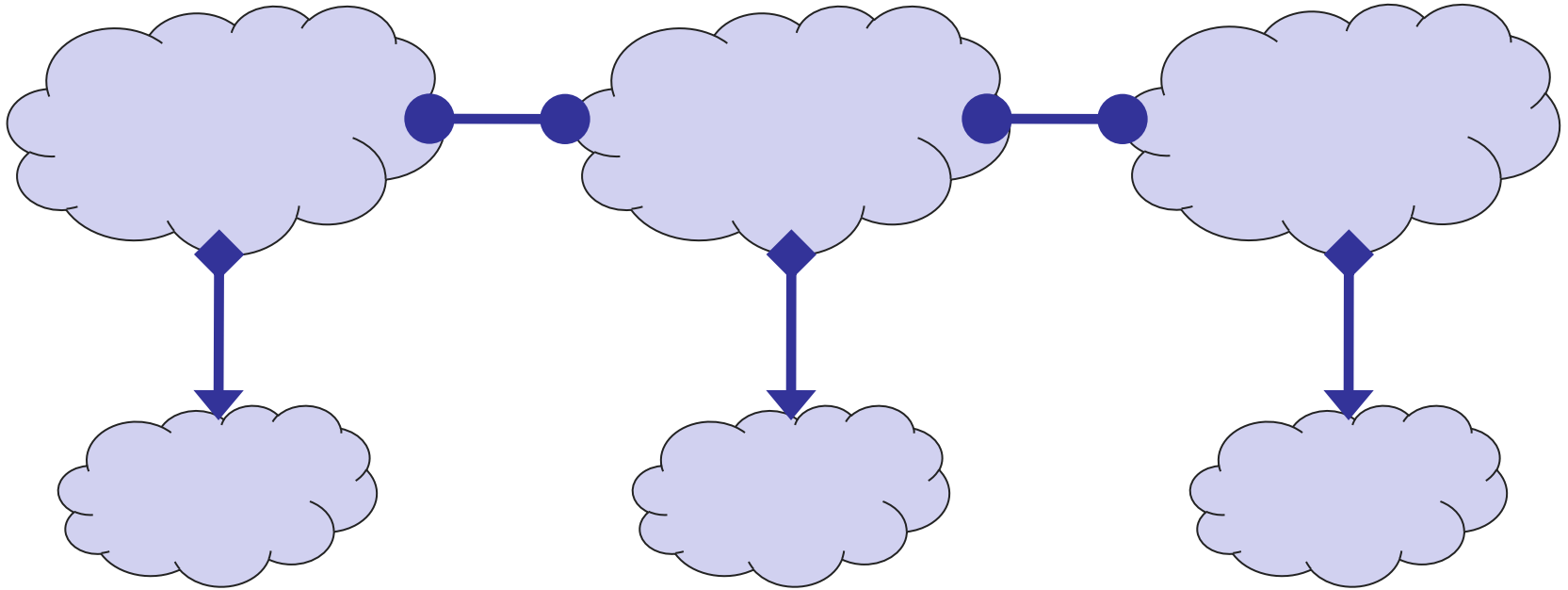
Administrative structure shapes Inter-domain routing

- ▣ ASes want freedom to pick routes based on policy
- ▣ ASes want autonomy
- ▣ ASes want privacy

Topology & policy shaped by inter-AS business relationship

- Three basic kinds of relationships between ASes
 - AS A can be AS B's customer
 - AS A can be AS B's provider
 - AS A can be AS B's peer
- Business implications
 - Customer pays provider
 - Peers don't pay each other
 - » Exchange roughly equal traffic

Business relationships



Relations between ASes

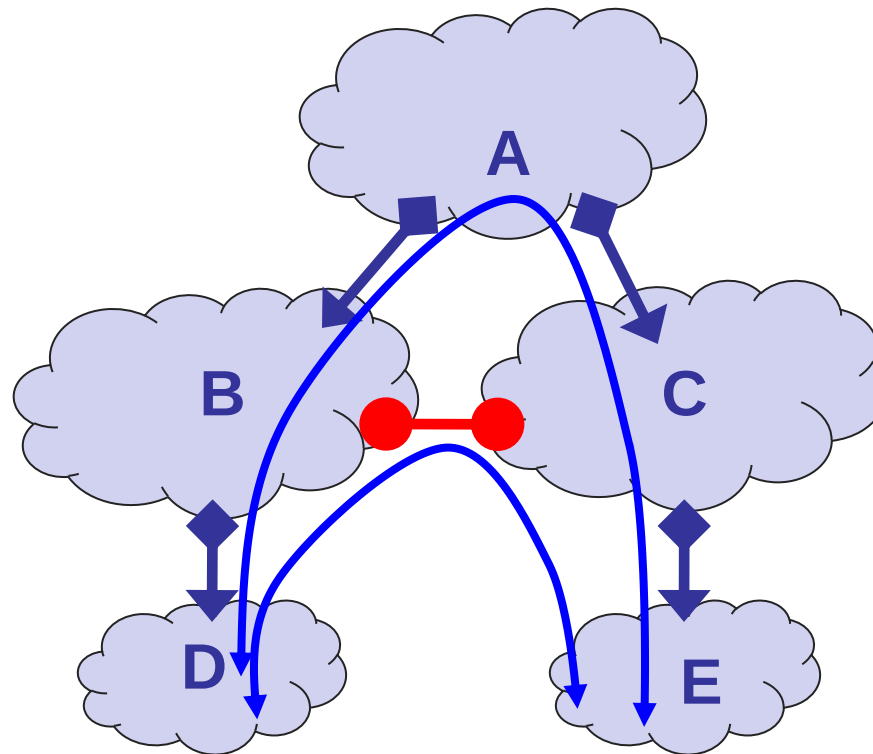
provider $\longleftrightarrow$ customer

peer $\bullet\text{---}\bullet$ peer

Business implications

- Customers pay provider
- Peers don't pay each other

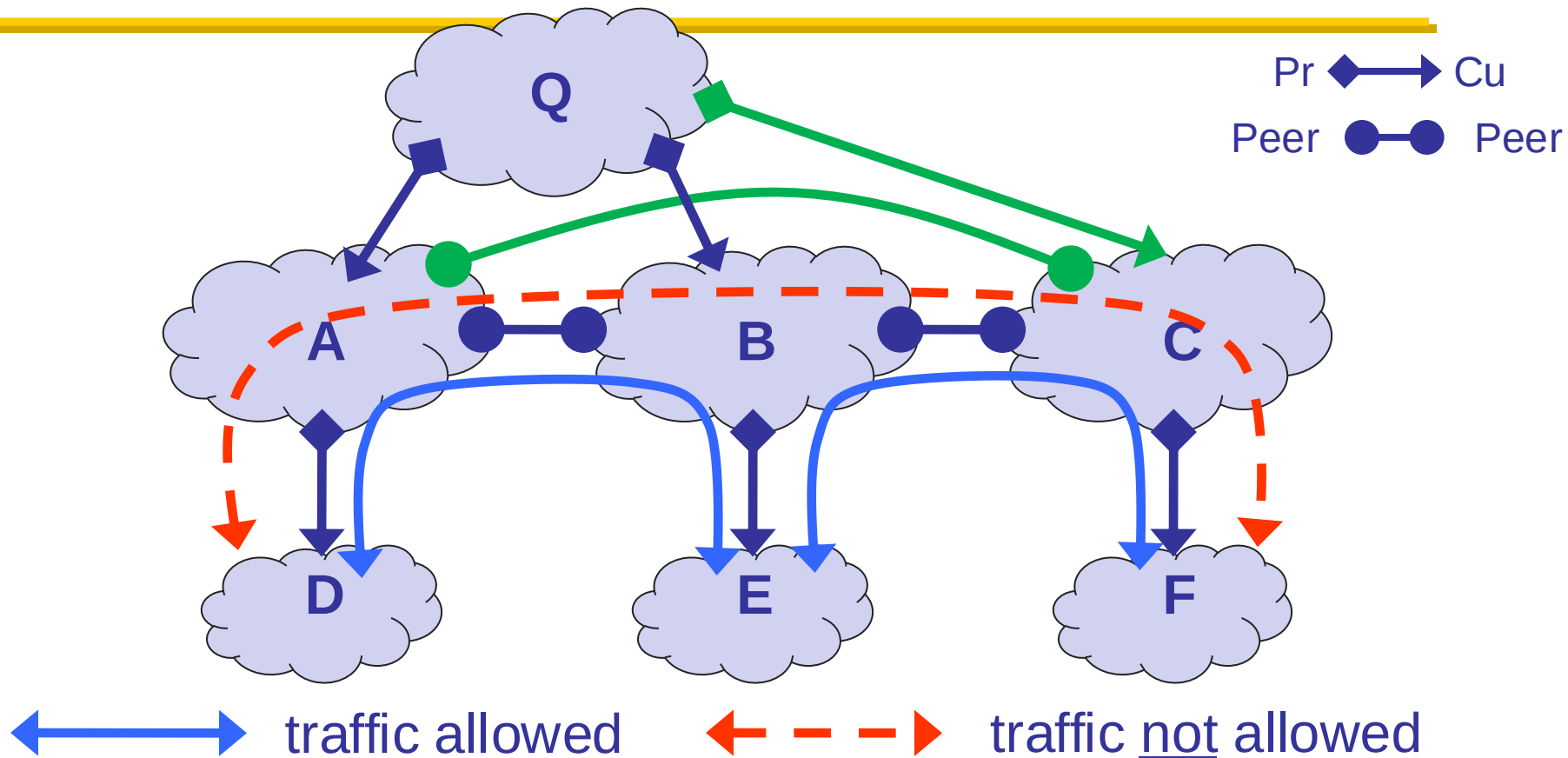
Why peer?



D and E
communicate a lot

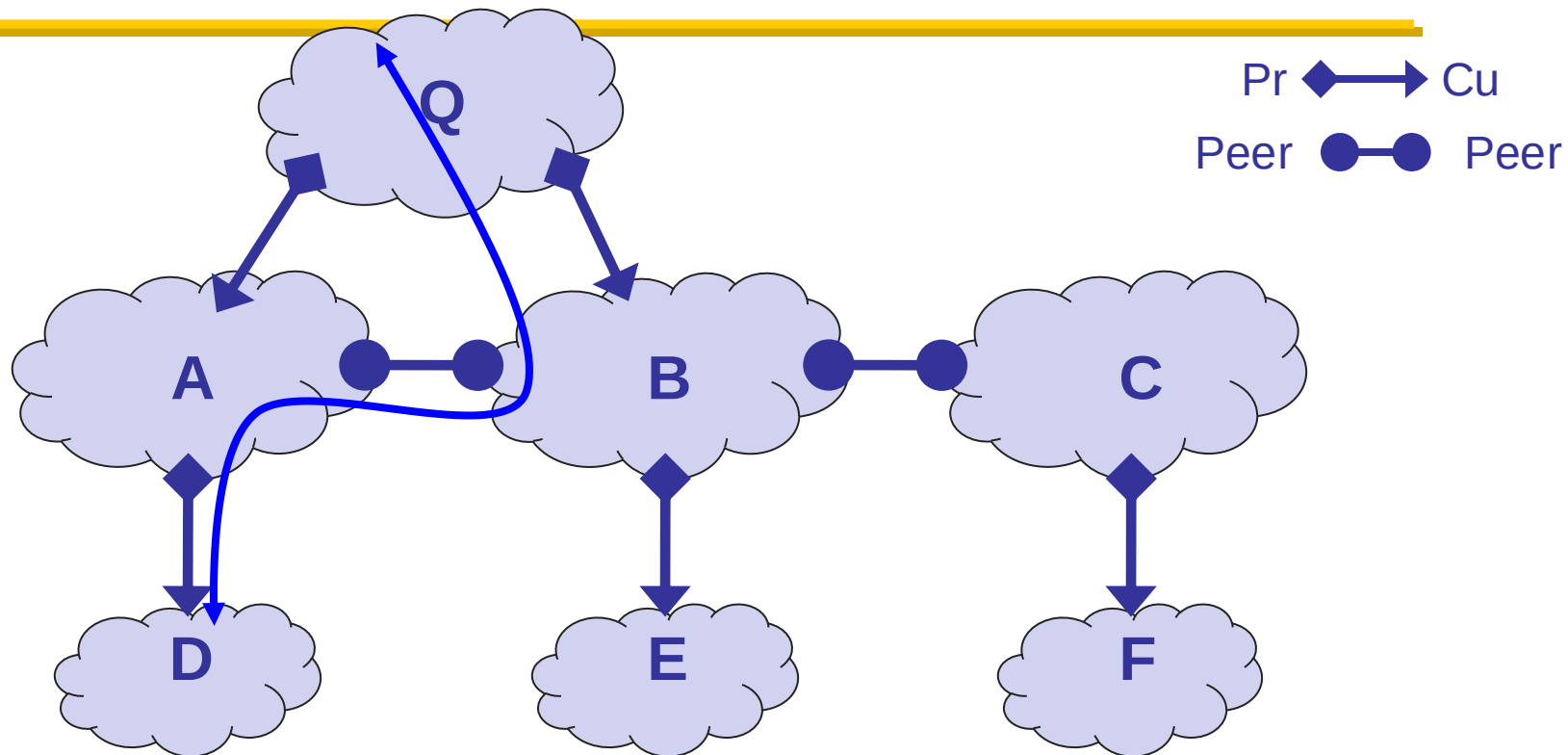
Peering saves
B and C money

Routing follows the money!



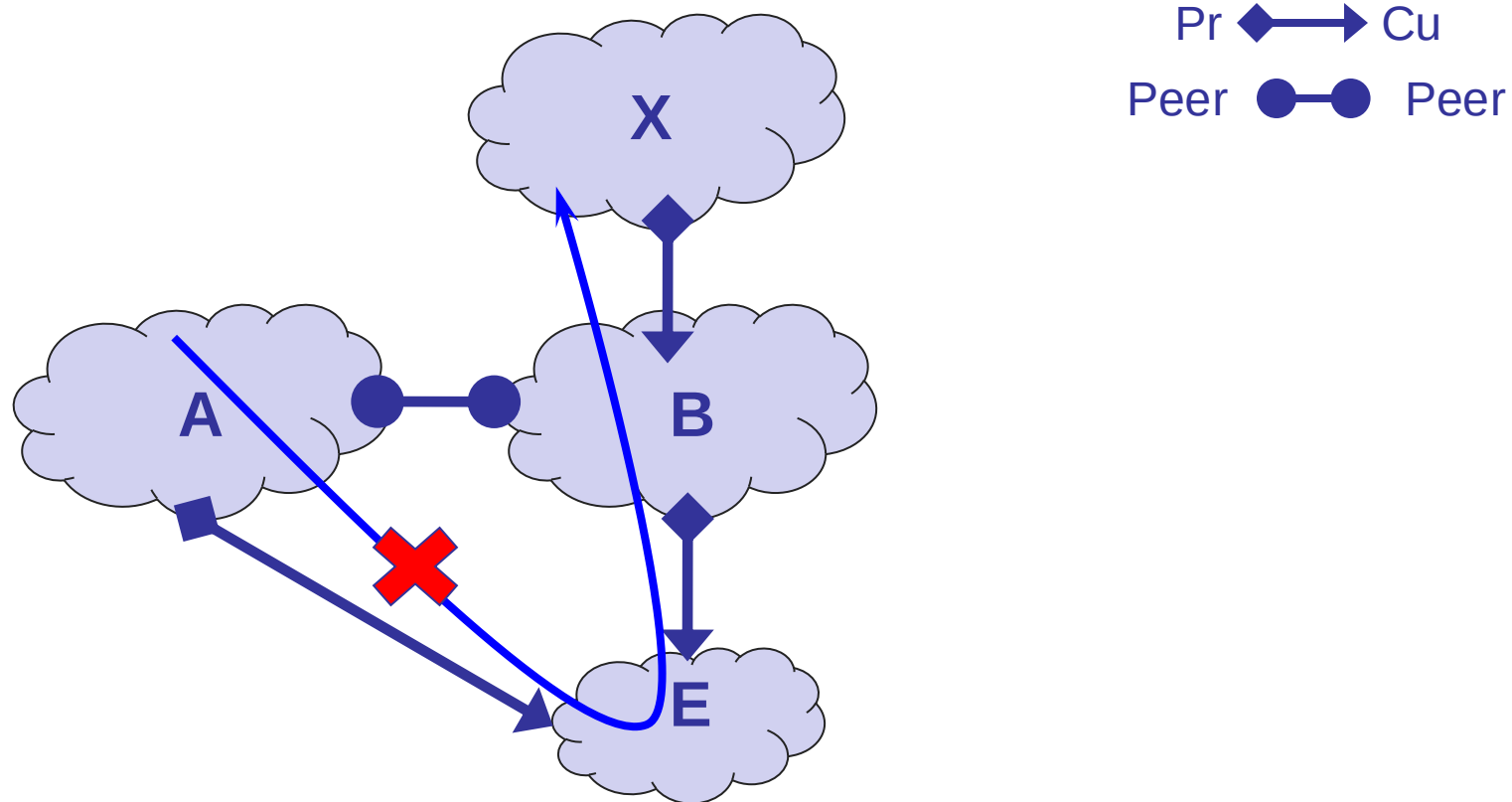
- ASes provide “transit” between their customers
- Peers do not provide transit between other peers

Routing follows the money!



- An AS only carries traffic to/from its own customers over a peering link

Routing follows the money!



- Routes are “valley” free (more details later)

In short

- AS topology reflects business relationships between ASes
- Business relationships between ASes impact which routes are acceptable

Summary

- Two key challenges in inter-domain routing
 - Scaling (Addressing)
 - Administrative structure (BGP)
 - » Issues of autonomy, policy, privacy

- **Next lecture:** BGP policies, protocol, and challenges